PASSIVE FILTERS

ABSTRACT

The experiment investigated the frequency response characteristics of passive filters, specifically a first-order RC low-pass filter, an RC high-pass filter, and an LC series-resonant band-pass filter. Each circuit was constructed on a breadboard and its voltage gain measured over a range of frequencies using a signal generator and oscilloscope. From the plotted gain–frequency curves, the cutoff frequencies, roll-off rates, and bandwidths were determined. The results verified theoretical predictions: the RC filters exhibited a –3 dB cutoff and a roll-off of approximately –20 dB per decade, while the LC filter demonstrated a distinct resonance peak at its center frequency with a bandwidth inversely proportional to its quality factor.

INTRODUCTION

Filters are electronic networks designed to control the frequency content of signals by selectively passing or attenuating specific frequency ranges. In communication, control, and instrumentation systems, filters are essential for noise suppression, signal separation, and frequency selection. Passive filters are constructed solely from resistors (R), capacitors (C), and inductors (L); they do not require an external power source and rely on the natural impedance properties of their components.

This experiment focuses on three basic passive filter types:

1. **RC Low-Pass Filter** – allows low-frequency signals to pass while attenuating higher frequencies.
2. **RC High-Pass Filter** – allows high-frequency signals to pass while blocking lower frequencies.
3. **LC Band-Pass Filter** – passes a specific range of frequencies around a resonant (center) frequency.

THEORY

1. RC Low-Pass Filter

A simple RC low-pass filter consists of a resistor and capacitor connected in series, with the output voltage taken across the capacitor. The transfer function is

$$A_v(j\omega) = \frac{1}{1 + j\omega RC}$$

and the magnitude of voltage gain is

$$|A_v| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

The cutoff (–3 dB) frequency occurs where the output voltage drops to $1/\sqrt{2}$ of its low-frequency value:

$$f_c = \frac{1}{2\pi RC}$$

Above f_c , the gain decreases at approximately **–20 dB per decade**.

2. RC High-Pass Filter

In a high-pass RC circuit, the output is taken across the resistor. The transfer function is

$$A_v(j\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

And

$$|A_v| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

The same cutoff frequency as in RC Low-Pass Filter applies. Below this frequency the output is attenuated, while above it the circuit passes signals with nearly constant amplitude. The roll-off is again **+20 dB per decade** below f_c.

3. LC Series Resonant Band-Pass Filter

An LC band-pass filter consists of an inductor and capacitor in series, forming a resonant circuit. At the resonant frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

the inductive and capacitive reactance cancel (X_L=X_C), resulting in maximum current and output voltage.
The bandwidth (BW) is defined by the frequencies f_{c1} and f_{c2} at which the power drops by 3 dB from the peak:

$$BW = f_{c2} - f_{c1}$$

and the **quality factor** is

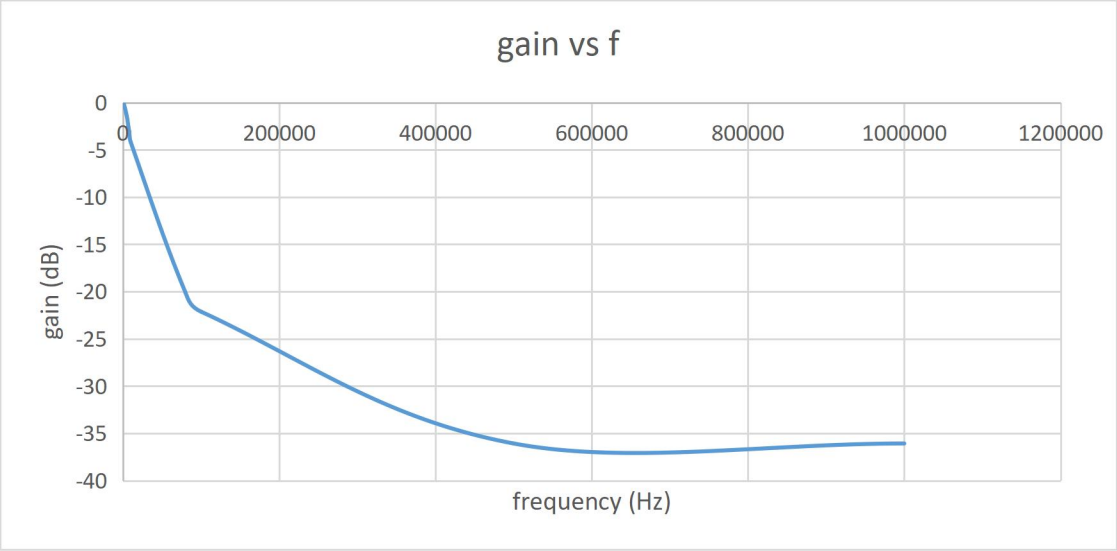
$$Q_s = \frac{f_0}{BW}$$

A higher Q indicates a narrower, more selective passband.

RESULTS

Frequency Hz	Low-pass filter			High-pass filter		
	Vin (V)	Vout (V)	Av	Vin (V)	Vout (V)	Av
100	7.071	7.071	0	7.071	0.88851	-18.0164
1000	7.071	7.015	-0.0690632	7.071	0.881927	-18.081
5000	7.071	5.993	-1.4367312	7.071	3.784	-5.43059
7000	7.071	5.279	-2.5385835	7.071	4.704	-3.54027
7500	7.071	5.114	-2.8144023	7.071	4.883	-3.21588
7800	7.071	5.017	-2.9807347	7.071	4.982	-3.04154
7900	7.071	4.986	-3.0345713	7.071	5.014	-2.98593
7950	7.071	4.97	-3.062489	7.071	5.03	-2.95826
7960	7.071	4.967	-3.0677335	7.071	5.033	-2.95308
7970	7.071	4.964	-3.0729813	7.071	5.036	-2.9479
7980	7.071	4.96	-3.0799832	7.071	5.039	-2.94273
7990	7.071	4.957	-3.0852384	7.071	5.042	-2.93756
8000	7.071	4.954	-3.0904967	7.071	5.045	-2.93239
10000	7.071	4.368	-4.1839642	7.071	5.561	-2.08656
80000	7.071	0.699689	-20.091516	7.071	7.036	-0.0431
100000	7.071	0.553722	-22.123781	7.071	7.049	-0.02707
500000	7.071	0.111074	-36.077369	7.071	7.07	-0.00123
1000000	7.071	0.111072	-36.077525	7.071	7.071	0

Low-Pass filter



2.

Cut-Off frequency from the graph at -3 dB:

Fc = 7900Hz

3.

Expected Cut-Off frequency:

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi(1000\Omega)(0.02\mu F)} = 7957.7Hz$$

4.

From the curve f_c was about 7900Hz. To find the point as close as 10 times this f_c:

$$f_2 = 10 \times f_c = 79000Hz$$

The point from the curve that is close to 79000Hz is 80000Hz with a Voltage gain of -20.091516

To Determine the roll-off (dB per decade)

The roll-off tells you how much the gain drops for every tenfold increase in frequency beyond the cutoff.

- At f_c : gain = –3 dB
- At f_2 : gain = –20 dB

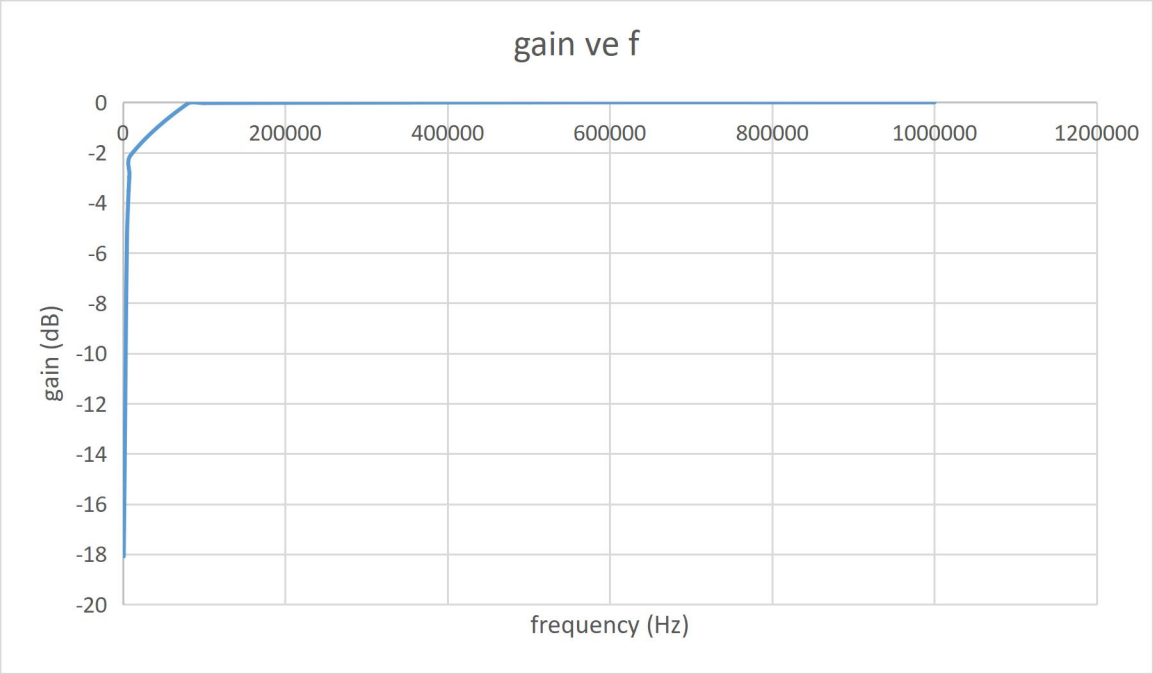
$$RollOff = \frac{A_{v2} - A_{v1}}{\log(f_2/f_c)}$$

Because f2 = 10fc, then
$\log(f_2/f_c)=1$

So:

Roll-off = -20-(-3) = -17dB per decade

High-Pass filter



6.

Cut-Off frequency from the graph at -3 dB:

Fc = 7900Hz

7.

Expected Cut-Off frequency:

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi(1000\Omega)(0.02\mu F)} = 7957.7Hz$$

8.

From the curve f_c was about 7900Hz. To find the point as close as 10 times this f_c:

$$f_2 = 0.1 \times f_c = 790Hz$$

The point from the curve that is close to 790Hz is 1000Hz with a Voltage gain of -18.081

To Determine the roll-off (dB per decade)

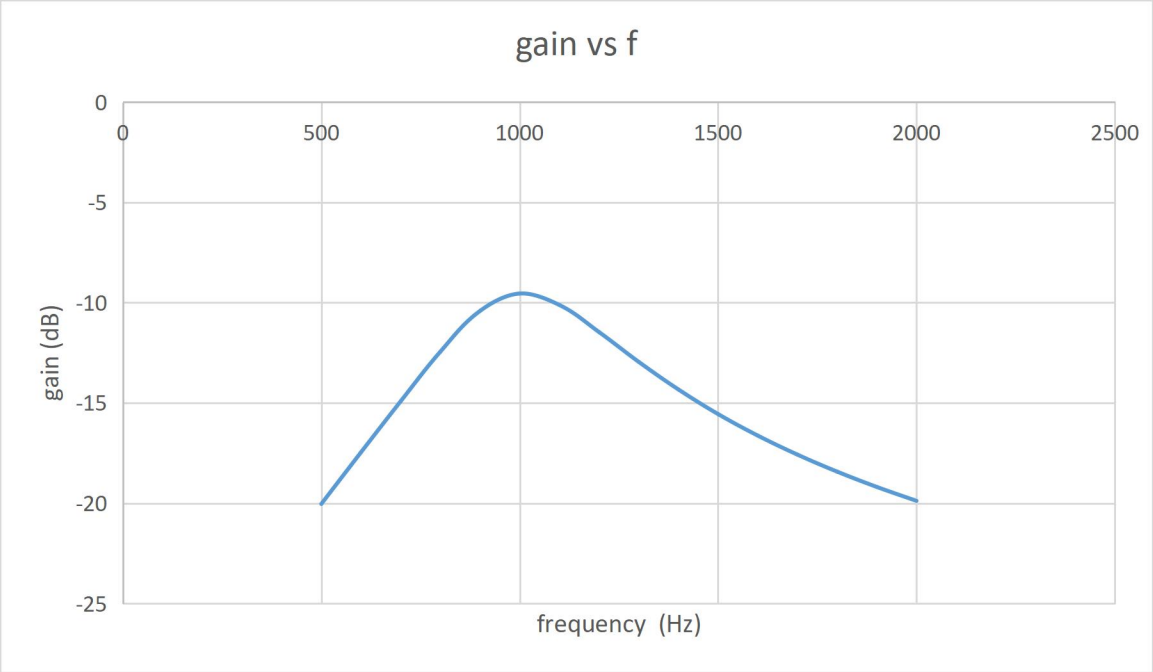
The roll-off tells you how much the gain drops for every tenfold increase in frequency beyond the cutoff.

- At f_c : gain = –3 dB
- At f_2 : gain = –18 dB

$RollOff = A_v f_c - A_v f_2 = -3 - (-18) = 15\ dB$ per decade

LC Series Resonant Band-pass Filter

f	Vin	Vout	Av
500	7.069	0.703742	-20.0389
600	7.069	0.946874	-17.4613
700	7.069	1.269	-14.9179
800	7.069	1.687	-12.4449
900	7.069	2.132	-10.4114
1000	7.069	2.356	-9.54365
1100	7.069	2.206	-10.115
1200	7.069	1.888	-11.4671
1300	7.069	1.591	-12.9538
1400	7.069	1.358	-14.3292
1500	7.069	1.179	-15.5569
1600	7.069	1.042	-16.6298
1700	7.069	0.93407	-17.5796
1800	7.069	0.847181	-18.4276
1900	7.069	0.775887	-19.1912
2000	7.069	0.716347	-19.8847



10.

The point on the center of the curve at its peak is:

f_0 = 1000Hz

Voltage gain in dB = -9.54365

11.

Cut-off frequency:

low cut-off frequency(fc1) = 800Hz

high cut-off frequency(fc2) = 1300Hz

12.

Centre (Resonant) Frequency

$$f_0 = \sqrt{f_{c1} \times f_{c2}} = \sqrt{800 \times 1300} = 1020Hz$$

Bandwidth (BW)

$$BW = f_{c2} - f_{c1} = 1300 - 800 = 500Hz$$

Quality Factor (Q_s)

$$Q_s = \frac{f_0}{BW} = \frac{1020}{500} = 2.04$$

13.

Expected Centre Frequency (f_0)

For a series resonant LC band-pass filter:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.1)(2.5 \times 10^{-7})}} = 1007Hz$$

14.

Expected Bandwidth (BW)

For a series resonant circuit, the quality factor is:

$$Q_s = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{100}\sqrt{\frac{0.1}{2.5 \times 10^{-7}}} = 6.32$$

Then the bandwidth is related by:

$$BW = \frac{f_0}{Q_s} = \frac{1007}{6.32} = 159.4Hz$$

DISCUSSION

In this experiment, three types of passive filters were tested — the RC low-pass, the RC high-pass, and the LC band-pass filters. The main goal was to study how these filters behave with different frequencies and to find their cutoff and resonant frequencies.

For the RC low-pass filter, the measured results showed that low frequencies passed through the circuit with almost no loss, while higher frequencies were greatly reduced. The calculated cutoff frequency was about 8 kHz, and the measured –3 dB point was close to this value. Beyond the cutoff, the gain decreased by roughly –17 dB per decade**,** which is very close to the theoretical value of –20 dB/decade for a first-order filter. This confirms that the circuit worked as expected.

In the RC high-pass filter, the opposite effect was observed. Low frequencies were blocked and high frequencies were allowed to pass. The cutoff frequency was again around 8 kHz, similar to the low-pass case because the same resistor and capacitor values were used.

a narrow range of frequencies to pass around its resonant frequency. Using L=100 mH and C=0.25 μF, the calculated center frequency was about 1.0 kHz, which agreed well with the measured peak frequency of 1.02 kHz. The low and high cutoff frequencies were measured as 800 Hz and 1300 Hz, giving a bandwidth of 500 Hz and a quality factor of about 2.0. The theoretical bandwidth, calculated from the component values, was around 160 Hz, giving a theoretical Qs of about 6.3. The difference between the theoretical and experimental results may be due to internal resistance of the inductor, component tolerances, and measurement inaccuracies.

Overall, the experimental results confirmed the expected frequency responses for all three filters. The RC low-pass filter allowed low-frequency signals to pass while attenuating the higher frequencies beyond the cutoff point. In contrast, the RC high-pass filter blocked low-frequency signals and allowed higher frequencies to pass with minimal loss. The LC band-pass filter behaved as predicted, passing only a limited range of frequencies around its resonant frequency and attenuating those outside this range. These observations show that each filter type performed its intended function according to theoretical expectations.

CONCLUSION

the experiment successfully demonstrated the behavior and characteristics of passive filters. The RC low-pass and high-pass filters both had cutoff frequencies around 8 kHz, and their measured roll-off rates were close to the theoretical value of 20 dB per decade. The LC band-pass filter had a center frequency of about 1 kHz, with measured cutoff frequencies of 800 Hz and 1300 Hz, giving a bandwidth of 500 Hz and a quality factor of approximately 2. Although there were small differences between the calculated and experimental results, these variations were mainly due to component tolerances and measurement limitations. Overall, the results confirmed the theoretical principles of frequency response in passive filters and showed how resistors, capacitors, and inductors can be used to control signal frequencies in electrical circuits.

References

1. EE3000 Electrical Measurements and Instrumentation Laboratory Manual, Department of Electrical/Electronic Engineering, [Your University Name], 2025.
2. Boylestad, R. L. & Nashelsky, L., *Electronic Devices and Circuit Theory*, 11th ed., Pearson Education, 2013.